

# Reconstructing the Barcelona Exhibition of 1929 through AI Based Generative 3D Reconstruction

Carles Pàmies

Escola Tècnica Superior d'Arquitectura de Barcelona ETSAB – Universitat Politècnica de Catalunya (UPC)  
Avinguda Diagonal, 649, 08028 Barcelona, Spain | carles.pamies.sauret@upc.edu

## Abstract

This article presents an innovative study of the 1929 International Exposition of Barcelona, focused on the virtual reconstruction of three vanished pavilions —Yugoslavia, Romania, and Hungary— through generative artificial intelligence (AI). The paper also constitutes a state of the art of generative 3D AI in 2026, providing a documented snapshot of the current capabilities of these tools in a rapidly evolving field, intended to serve as a comparative reference as these technologies advance in the near future. Four 3D-from-2D AI tools —TripoAI, Hunyuan 2.5, Hitem3D, and Meshy— are systematically evaluated for geometric accuracy, historical fidelity, texture quality, render time, and educational and museographic applicability. Results reveal significant differences among the tools: Meshy and Hunyuan 2.5 offer the most promising outcomes for heritage preservation applications, while TripoAI and Hitem3D show notable limitations in geometric precision. The study establishes a methodological precedent for the digital recovery of ephemeral architectural heritage and raises ethical and epistemological considerations regarding AI-driven historical reconstruction.

**Keywords:** Architecture, Digital Reconstruction, Heritage, Generative Artificial Intelligence, State of the Art 2026, Urban Planning.

## 1. INTRODUCTION

The 1929 International Exposition of Barcelona was a transformative milestone that left a profound mark on the city, representing a turning point in its urban and cultural development. The event drove a comprehensive renewal of Montjuïc and its surroundings, providing Barcelona with new infrastructure that consolidated its image as a modern and cosmopolitan city on the international stage.

In architectural terms, the Exposition served as a laboratory for the emerging styles of the early twentieth century, fostering the introduction of international avant-garde movements into Spain. Iconic works such as Ludwig Mies van der Rohe's German Pavilion —with its austere formal purity and innovative use of materials— anticipated aesthetic currents that would define the following decades.

This article aims primarily to analyse the potential of generative artificial intelligence in the 3D reconstruction of vanished architectural heritage, taking as a case study three international pavilions from the Exposition. The research seeks to address the following question: to what extent can AI contribute to a deeper, more interactive, and more critical understanding of ephemeral architectural heritage?

Beyond this specific case, the paper constitutes a state of the art of generative 3D AI in 2026 —a documented snapshot of the current capabilities of these tools in a field that is evolving with extraordinary speed. The rapid pace at which 3D generators are launched and improved makes this comparative analysis a potentially valuable reference for future research assessing the progress achieved once these technologies have advanced further. In this sense, the article aspires not only to resolve the concrete case of the 1929 pavilions, but also to provide a rigorous record of the state of AI-driven 3D generation at the time of its publication.

## 2. THEORETICAL FRAMEWORK: DIGITAL HERITAGE AND TECHNOLOGICAL INNOVATION

### 3. THE 1929 INTERNATIONAL EXPOSITION OF BARCELONA

The 1929 International Exposition of Barcelona represented a pivotal moment in the city's process of urban modernisation, acting as a mirror of the political, social, and urban tensions of the period. The origins of the event can be traced to an article published in 1905 by Josep Puig i Cadafalch, who proposed a new universal exposition as an instrument for promoting Barcelona's transition towards modernity.

The selection of Montjuïc mountain as the site —over alternatives such as the Plaça de les Glòries Catalanes— was contested due to its challenging topography, yet ultimately prevailed as a bold wager on the city's future projection. Construction works and landscaping unfolded between 1917 and 1929, culminating in the official inauguration on 19 May 1929.

The Exposition brought together a remarkable international representation. Among the most significant pavilions: the German Pavilion by Ludwig Mies van der Rohe and Lilly Reich —demolished in 1930 and reconstructed in 1986—, the French Pavilion by Georges Wybo, the Italian by Piero Portaluppi, the Hungarian by Dénes Györgyi and Nikolaus Menyhért, the Romanian by Duiliu Marcu, and the Yugoslav by Dragiša Brašovan. Following the event, numerous pavilions were demolished, while the principal buildings were repurposed for cultural and recreational uses (Figure 1).



Figure 1. The Exposition in 1929 (Source: Arxiu Municipal Contemporani)

### 4. METHODOLOGY

The study combines historical and documentary analysis with 3D generation using generative AI. Primary sources include archival historical photographs, contemporary chronicles, and, where available, original architectural plans. Four generative AI tools for converting 2D images into 3D models were applied under different input conditions in order to compare their performance.

The evaluation criteria were: geometric fidelity, historical and stylistic coherence, texture quality, render time, polygon density, and applicability in educational and museographic contexts.

#### 4.1 Selection and Justification of the Case-Study Pavilions

The selection of the three pavilions under study responds to well-defined and complementary criteria:

- **Yugoslav Pavilion (Dragiša Brašovan):** Selected for its formal singularity —a star-shaped plan of avant-garde character— and for being the best documented of the three (four photographs are available: front, rear, and two side views), allowing a more reliable comparison of AI tool outputs.
- **Romanian Pavilion (Duiliu Marcu):** Chosen for its representativeness of historicist eclecticism with Transylvanian influences, and for presenting an intermediate level of graphic documentation (four photographs from various frontal and lateral angles). It enables assessment of generator performance when confronted with highly ornamented architecture.
- **Hungarian Pavilion (Dénes Györgyi and Nikolaus Menyhért):** Included precisely because it represents the case of most limited documentary availability —a single oblique frontal photograph— allowing evaluation of tool behaviour under minimal input conditions, a scenario frequently encountered in the recovery of vanished architectural heritage.

The combination of these three cases deliberately provides a diverse spectrum: stylistic diversity (expressionist rationalism, historicist eclecticism), documentary diversity (from four images to a single

photograph), and geometric complexity diversity. This variety maximises the comparative value of the analysis (Figure 2).



Figure 2. The Pavilions of Yugoslavia, Rumania and Hungría (Source: Arxiu Amatller)

## 4.2 Data Sources for Reconstruction

The virtual reconstruction is grounded in the systematic compilation and analysis of historical sources of different types:

- **Historical photographs:** Period images from both official and personal archives, essential for interpreting textures, materials, and environmental context. They constitute the primary input for the generative AI tools (Figure 3).
- **Written descriptions and contemporary chronicles:** Press articles, memoirs, and official guides providing qualitative information on materials, colours, and details not always visible in photographs.
- **Architectural plans (where available):** Used as dimensional verification references, though their availability for the three selected pavilions is only partial.

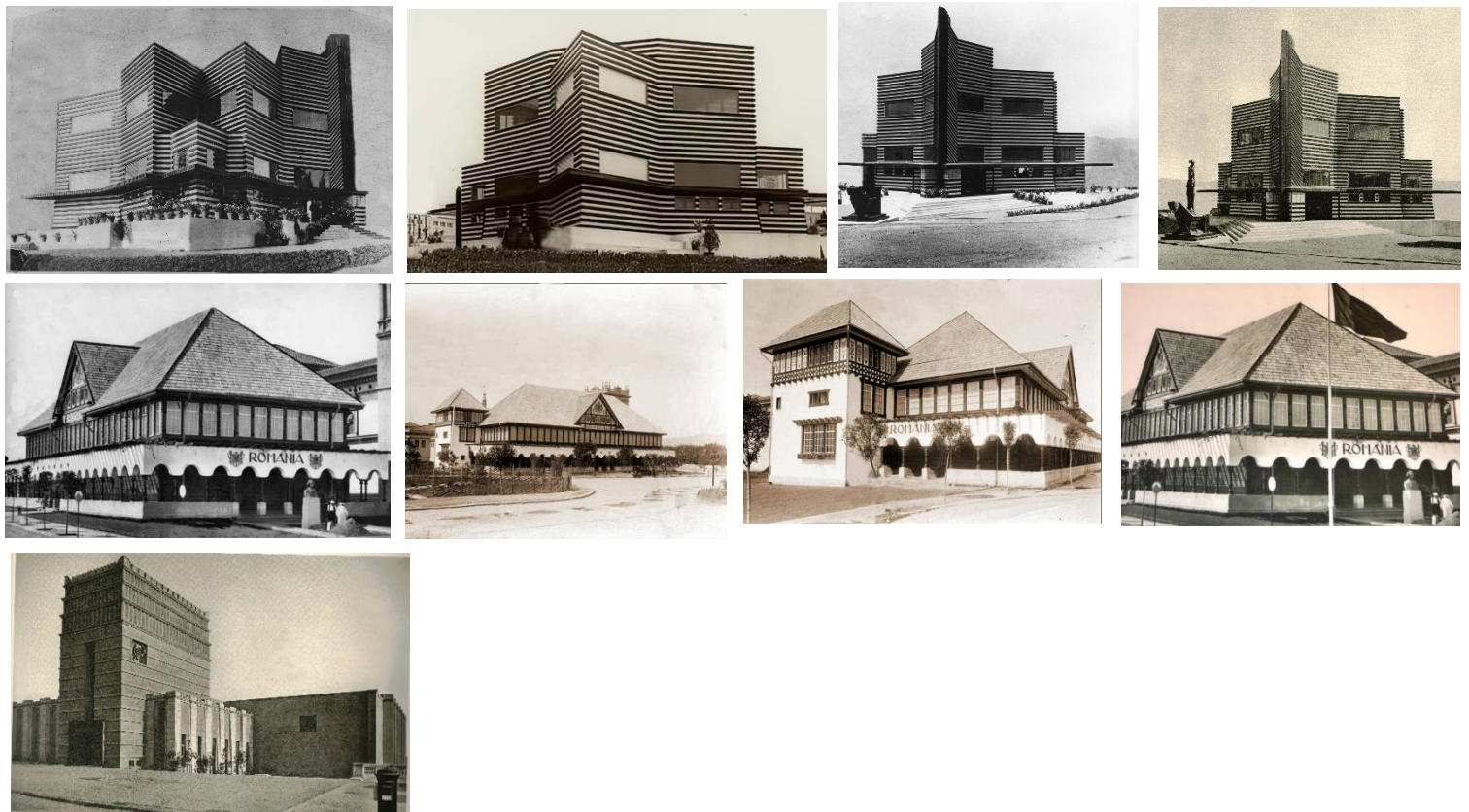


Figure 3. Images used for generate Pavilions of: Yugoslavia (first row) , Rumania (second row) and Hungría (third row). (Source: Arxiu Amatller)

### 4.3 Generative 3D AI Tools: State of the Art 2026

The four tools selected represent the state of the art in 3D-from-2D generation in 2026. They were chosen for their public availability, the variety of technical approaches they embody, and their capacity to process complex visual inputs with historical and stylistic references. Their principal characteristics are described below.

#### ***TripoAI (Tripo AI) — VAST***

Emerging from an open-source collaboration in 2025 with a public launch in September of that year, TripoAI generates 3D models in approximately 10 seconds from text, single images, or multiple photographs. Based on scalable diffusion models, it offers pixel-accurate topology. Multiple photographs from different angles are supported for enhanced precision. Export formats: GLB, FBX, OBJ, USD, STL.

#### ***Hunyuan 2.5 (Hunyuan 3D 2.5) — Tencent AI Lab***

An evolution of Hunyuan3D (2024), launched in April 2025 as a fully open-source large-scale 3D synthesis system. It incorporates Physics-Based Rendering (PBR) texture synthesis, significantly improving material fidelity. Optional multi-view support is available for increased accuracy. Generation time: seconds to minutes. Formats: OBJ, GLB. Available free for basic use and integrable with ComfyUI.

#### ***Hitem3D — Math Magic***

Launched in July 2025 as the first AI 3D generator with ultra-high resolution (up to 1536). The v1.5 update of October 2025 introduced multi-view support. Converts photographs, sketches, or concept art into textured 3D models in approximately 150 seconds. Supports 2–4 input images, with a simple background recommended. Formats: OBJ, GLB, STL.

#### ***Meshy — Meshy.ai***

A platform active since 2024, with Meshy 6 launched in October 2025 and primarily aimed at game creators and designers. Converts 2D images into detailed 3D models in approximately 1 minute. Multi-view generation analyses multiple perspectives (front, side, rear), and integrated animation and refinement tools are included. Particularly notable for texture quality. Formats: GLB, OBJ, FBX.

## 5. RESULTS

The four generative AI tools were applied to the three pavilions under different input conditions, reflecting the real documentary availability of each case:

- Yugoslav Pavilion: 4 photographs (front, rear, and two side views).
- Romanian Pavilion: 4 photographs (3 front-lateral and 1 rear).
- Hungarian Pavilion: 1 photograph (oblique frontal view).

This deliberate variation in the number of input images allows evaluation of tool performance not only under optimal conditions but also under documentary constraints —a scenario that is common in the recovery of vanished heritage.

### 5.1 Quantitative Data: Render Times and Polygon Density

Table 1 presents the quantitative results of the twelve pavilion–tool combinations evaluated:

| Pavilion   | Tool        | Render time | Polygons |
|------------|-------------|-------------|----------|
| YUGOSLAVIA | TripoAI     | 2' 15"      | 151 K    |
| YUGOSLAVIA | Hunyuan 2.5 | 4' 51"      | 1.5 M    |
| YUGOSLAVIA | Hitem3D     | 22'         | 1.9 M    |
| YUGOSLAVIA | Meshy       | 4' 30"      | 1.2 M    |
| ROMANIA    | TripoAI     | 1'          | 144 K    |

| Pavilion | Tool        | Render time | Polygons |
|----------|-------------|-------------|----------|
| ROMANIA  | Hunyuan 2.5 | 6'          | 1.5 M    |
| ROMANIA  | Hitem3D     | 21'         | 1.9 M    |
| ROMANIA  | Meshy       | 5' 30"      | 961 K    |
| HUNGARY  | TripoAI     | 3'          | 1 M      |
| HUNGARY  | Hunyuan 2.5 | 4'          | 1.5 M    |
| HUNGARY  | Hitem3D     | 22'         | 1.9 M    |
| HUNGARY  | Meshy       | 5'          | 496 K    |

Table 1. Tools applied, render times, and polygon density per pavilion.

## 5.2 Qualitative Results by Tool

The results reveal significant differences among the evaluated tools in terms of geometric fidelity, texture quality, and suitability for heritage applications:

**TripoAI:** Produces low geometric precision meshes across all three pavilions. Polygons form irregular volumes that evoke the general form but lack the detail required for technical applications. It is the fastest tool (1–3 minutes) and generates the lowest polygon counts (144K–1M). Suitable only for conceptual visualisations or preliminary explorations.(Figures 5, 7, 9)

**Hunyuan 2.5:** Delivers consistent and balanced results across all three pavilions, with ordered topology and moderate density (a constant 1.5M polygons). The generated form is reasonably adequate for digital reconstructions or 3D printing prototypes, though the polygon count may prove excessive for systems prioritising computational optimisation. The tool generates multiple variations, from which the two most historically coherent are selected.(Figures 5, 7, 9)

**Hitem3D:** Despite recording the highest constant polygon density (1.9M) and the longest render times (21–22 minutes), it produces the most disappointing results in terms of geometric coherence. The generated meshes are neither adequately formed nor fit for historical reconstructions, prototypes, or 3D printing applications.(Figures 5, 7, 9)

**Meshy:** Stands out as the most suitable option for heritage and educational applications. It generates meshes with clean topology, high stylistic fidelity, and notable quality in ornamental details and historical proportions. Its texturisation is markedly superior to that of the other three generators, even inferring plausible colours and materials not directly visible in the input photographs. Polygon density varies more across pavilions (496K–1.2M), adapting better to the complexity of each model.(Figures 5, 7, 9)

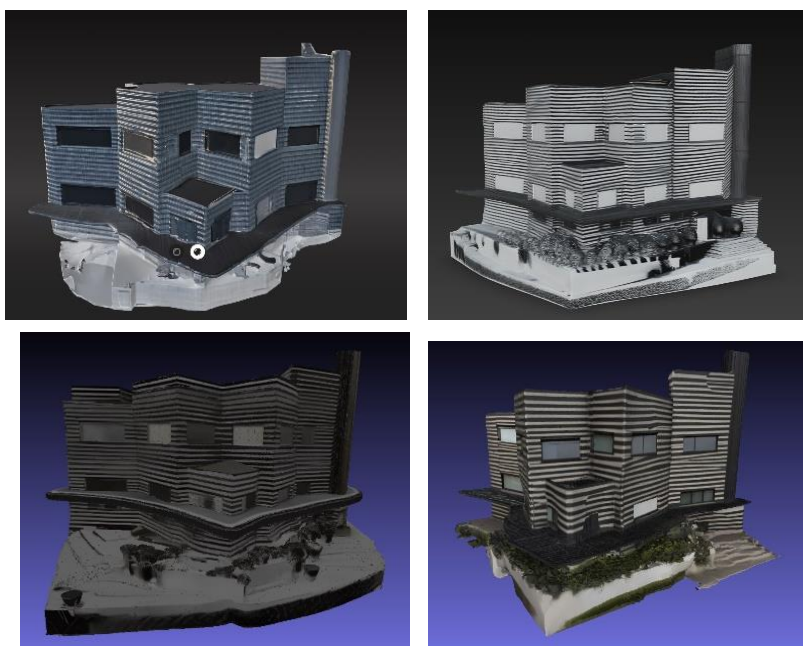


Figure 4. Renders of Pavilion of Yugoslavia: TripoAI, Hunyuan, Hitem3D y Meshy

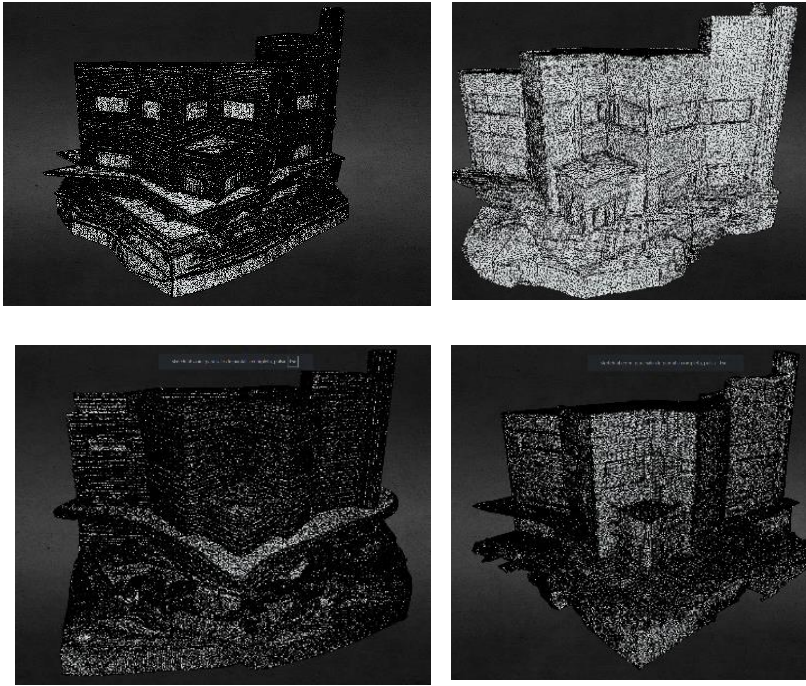


Figure 5. Meshes of Pavilion of Yugoslavia: Trippo, Hunyuan, Hitem3D y Meshy



Figure 6. Renders of Pavilion of Rumania: Trippo, Hunyuan, Hitem3D y Meshy

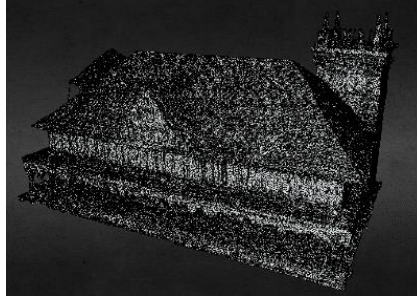
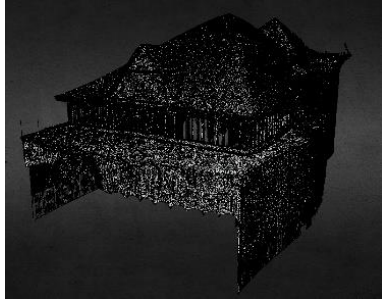
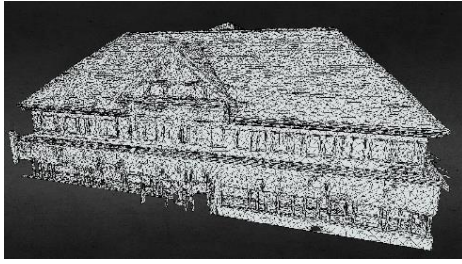


Figure 7. Meshes of Pavilion of Rumania: Trippo, Hunyuan, Hitem3D y Meshy

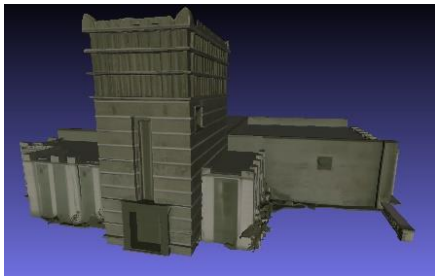


Figure 8. Renders of Pavilion of Hungría: Trippo, Hunyuan, Hitem3D y Meshy

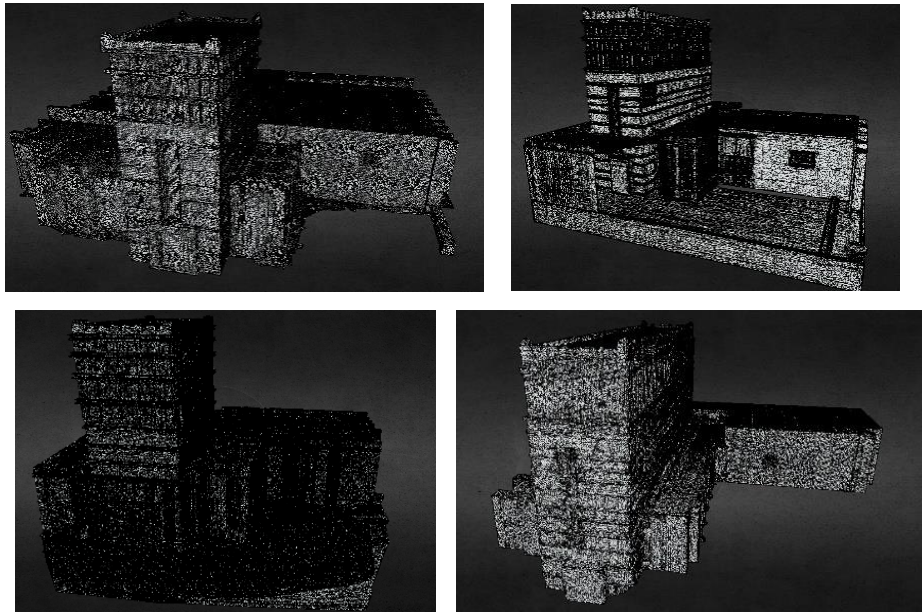


Figure 9. Meshes of Pavilion of Hungría: Tripo, Hunyuan, Hitem3D y Meshy

### 5.3 Qualitative Results by Pavilion

**Yugoslav Pavilion (best documented):** The availability of four photographs from different angles allows all tools to better capture the general geometry of the building. Meshy achieves a notably faithful reproduction of Braşovan's characteristic star-shaped plan and the ornamental details of the façade. Hunyuan produces geometrically plausible variations. TripoAI and Hitem3D generate recognisable but imprecise volumes. (Figures 4, 6, 8)

**Romanian Pavilion (intermediate documentation):** The historicist ornamentation with Transylvanian influences poses a greater challenge for the generators. Meshy better captures the decorative richness thanks to its textural superiority. Hunyuan produces acceptable geometry but loses ornamental detail. TripoAI generates the lowest density mesh (144K), insufficient for this type of architecture. (Figures 4, 6, 8)

**Hungarian Pavilion (minimal documentation):** With only a single oblique frontal photograph available, this case tests the inferential capacity of the tools at their limits. Results are overall poorer, confirming the direct correlation between documentary richness and output quality. Even so, Meshy and Hunyuan produce geometrically reasonable approximations that may serve as a starting point for an expert-supervised hypothetical reconstruction. (Figures 4, 6, 8)

The complete results —renders and meshes of the three pavilions for each tool— are available in the Sketchfab album: <https://skfb.ly/pCQpW>

## 6. DISCUSSION

The comparative evaluation of the four generative 3D AI tools confirms both the potential and the current limitations of these technologies for the preservation of vanished architectural heritage.

### 6.1 Systematic Comparison of Tools

Table 2 synthesises the qualitative assessment of the four tools according to the established criteria:

| Criterion           | TripoAI    | Hunyuan 2.5 | Hitem3D    | Meshy     |
|---------------------|------------|-------------|------------|-----------|
| Geometric precision | Low        | Medium-high | Low-medium | High      |
| Stylistic fidelity  | Low        | Medium      | Low        | High      |
| Texture quality     | Medium-low | Medium      | Medium     | Very high |

| Criterion                 | TripoAI          | Hunyuan 2.5     | Hitem3D       | Meshy                |
|---------------------------|------------------|-----------------|---------------|----------------------|
| Render time               | Very fast (1–3') | Fast (4–6')     | Slow (21–22') | Fast (4–5')          |
| Polygon density           | Low (144K–1M)    | Med-high (1.5M) | High (1.9M)   | Variable (496K–1.2M) |
| Heritage applicability    | Low              | Medium-high     | Low           | High                 |
| Educational applicability | Low              | Medium          | Low           | High                 |
| Overall assessment        | ★★☆☆☆            | ★★★★☆           | ★★☆☆☆         | ★★★★★                |

Table 2. Comparative assessment of the four generative 3D AI tools.

## 6.2 Methodological Implications

The findings confirm the potential of generative AI for architectural recreation, while also revealing limitations in topological precision and historical coherence. AI tools must be complemented by expert human review to guarantee heritage veracity. Output quality is directly correlated with the richness of input documentation: the Yugoslav Pavilion, with four photographs, yields substantially better results than the Hungarian, with a single image.

It is noteworthy that Hitem3D, despite being the generator with the highest processing time and the highest constant polygon density, delivers the poorest qualitative results. This highlights that polygon count is not synonymous with geometric quality or historical fidelity. Similarly, TripoAI's computational efficiency does not compensate for its geometric limitations in demanding heritage applications.

From a state-of-the-art perspective, it is important to stress that none of the four tools produces results directly usable without subsequent human intervention. This suggests that, in 2026, generative 3D AI should be conceived primarily as a tool for accelerating the reconstruction process—not for replacing expert judgement—and is most valuable in the hypothesis and formal exploration phases.

## 6.3 Ethical and Epistemological Considerations

The research raises relevant ethical and epistemological questions regarding the digital reconstruction of heritage: where is the boundary between faithful historical recreation and artistic interpretation? When does a reconstruction cease to be a reconstruction and become a reinterpretation? Meshy's remarkable capacity to infer colours and materials not visible in the input photographs—which may be regarded both as an asset and a risk to historical fidelity—illustrates this tension clearly.

Following the principles outlined by Hermon and Niccolucci (2020) and the recommendations of UNESCO (2019), all digital heritage reconstructions should be accompanied by metadata explicitly indicating the degree of historical certainty of each element of the model, distinguishing between what is documented, what is inferred, and what is hypothetical.

## 7. CONCLUSIONS

The study demonstrates that generative artificial intelligence provides powerful tools for the preservation and dissemination of architectural heritage, particularly in cases where physical documentation is limited. The results support the following conclusions:

- Among the evaluated tools, Meshy presents the most consistent and highest-quality results for heritage and educational purposes, excelling particularly in texturisation and stylistic fidelity. Hunyuan 2.5 is the second most recommendable option, offering a sound balance between geometric quality and efficiency.
- TripoAI and Hitem3D exhibit significant limitations that render them unsuitable for heritage preservation applications requiring historical rigour or technical precision.
- Documentary availability is the most determining factor in output quality: the greater the number and variety of angle photographs, the higher the geometric fidelity. This underscores the indispensable role of prior historical research as the foundation of digital reconstruction.
- At a methodological level, the study highlights the need to develop standardised criteria for evaluating the quality of digital heritage reconstructions, so that they may be rigorously integrated into professional, academic, and museographic applications.

- The present work constitutes a documented state of the art of AI-driven 3D generation in 2026. Given the accelerated pace of evolution in this field, this comparative analysis may serve as a reference for future research assessing the progress achieved —a benchmark against which the capabilities of tomorrow's tools may be measured.
- At a prospective level, the integration of multimodal AI, virtual reality, and digital twins holds the potential to transform the way we interact with heritage, offering immersive experiences that connect past and present in environments of digital museography and virtual tourism.

In sum, these AI tools mark notable advances in automated 3D modelling, yet still require improvements in historical precision, topological control, and scalability to meet the demands of real-world architectural applications. Future development should focus on enhancing reliability, multidisciplinary versatility, and integration with professional workflows, consolidating AI's role as an ally —not a substitute— of the expert in the digital preservation of architectural heritage.

## Declarations

### Data availability statement

The datasets generated and/or analyzed during the current study are not publicly available due to private rights in the ownership of the images.

### Funding

This research received no external funding.

### Clinical Trial Number

Not applicable

### Ethical approval

Not applicable.

### Consent to participate

Not applicable.

### Consent to publish

Not applicable.

## REFERENCES

- Addison, A. (2008). Emerging trends in virtual heritage. *IEEE Multimedia*. <https://doi.org/10.1109/93.848421>
- Cameron, F. (2010). Digital Futures I: Museum Collections, Digital Technologies, and the Cultural Construction of Knowledge. *Curator: The Museum Journal*, 50(3). <https://doi.org/10.1111/j.2151-6952.2003.tb00098.x>
- Corona-Figueroa, A. (2024). Repeat and Concatenate: 2D to 3D Image Translation. *IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops (CVPRW)*. <https://doi.org/10.1109/CVPRW63382.2024.00237>
- Debim, T., & Mulahusic, A. (2023). 3D Digital Reconstruction of Heritage Buildings by Using Old 2D Plans. *Lecture Notes in Networks and Systems*, 707.
- Europa Nostra. (2023). *Heritage in Action: European Heritage Award*. Europa Nostra Publications.
- Grandas, C. (2020). L'Exposició Internacional de Barcelona i la modernització de Barcelona. In: *Barcelona, capital de la radio*. ISBN 978-84-18199-63-9.
- Hermon, S., & Niccolucci, F. (2020). Principles of 3D Cultural Heritage Documentation. *Journal of Cultural Heritage*.

Hunyuan3D 2.1: From Images to High Fidelity 3D Assets with Production-Ready PBR. (2025). Tencent AI Lab. <https://doi.org/10.48550/arXiv.2506.15442>

Lin, J. (2025). Kiss3DGen: Repurposing Image Diffusion Models for 3D Asset Generation. IEEE/CVF Conference on Computer Vision and Pattern Recognition. <https://doi.org/10.1109/CVPR52734.2025.00551>

Mamand, S. S. (2024). 2D to 3D Image Conversion Algorithms. ITM Web of Conferences, 64. <https://doi.org/10.1051/itmconf/20246401010>

McMillan, E. (2025). From 2D to 3D: Deep Learning-Based Shape Reconstruction.

Nurgazy, T., Amirkhanova, B., et al. (2025). Using Artificial Intelligence to Create 3D Models of Manufacturing Equipment. <https://doi.org/10.1117/12.3065594>

Obsieger, B. (2003). Computer Aided Design and Manufacturing of Complex Parts. Sveuciliste u Rijeci.

Reddy, P. (2024). G3DR: Generative 3D Reconstruction in ImageNet. IEEE/CVF Conference on Computer Vision and Pattern Recognition. <https://doi.org/10.1109/CVPR52733.2024.00922>

Reilly, P. (1990). Towards a Virtual Archaeology. In: Rahtz, S. & Richards, J. (Eds.), Computer Applications in Archaeology.

Sarcevic, I., Doric, T., & Ivancevic, A. (2024). Capabilities of AI in Text-to-3D Model Generators for Mechanical Components Design. International Symposium on Graphic Engineering and Design. <https://doi.org/10.24867/GRID-2024-p44>

Saunders, B. (2000). From 2D to 3D: Numerical Grid Generation and the Visualization of Complex Surfaces. National Institute of Standards and Technology.

Solà-Morales, I. (1985). La Exposición Internacional de Barcelona 1914–1929: Arquitectura y Ciudad. Ed. Feria de Barcelona. ISBN: 84-398-3446-2.

Tang, J. (2024). LGM: Large Multi-View Gaussian Model for High Resolution 3D Content Creation. European Conference on Computer Vision. [https://doi.org/10.1007/978-3-031-73235-5\\_1](https://doi.org/10.1007/978-3-031-73235-5_1)

Tencent Hunyuan3D. (2025). Hunyuan3D Studio: End-to-end AI Pipeline for Game-Ready 3D Asset Generation. <https://doi.org/10.48550/arXiv.2509.12815>

Tochilkin, D., Pankratz, D., Liu, Z., & Huang, Z. (2024). TripoSR: Fast 3D Object Reconstruction from a Single Image. <https://doi.org/10.48550/arXiv.2403.02151>

UNESCO. (2019). Recommendation Concerning the Preservation of, and Access to, Documentary Heritage Including in Digital Form. UNESCO Publications.

Wang, T. (2022). Rodin: A Generative Model for Sculpting 3D Digital Avatars Using Diffusion. IEEE/CVF Conference on Computer Vision and Pattern Recognition. <https://doi.org/10.1109/CVPR52729.2023.00443>

Wu, T. (2025). Amodal3R: Amodal 3D Reconstruction from Occluded 2D Images. <https://doi.org/10.48550/arXiv.2503.13439>

Xiang, J. (2025). Structured 3D Latents for Scalable and Versatile 3D Generation. <https://doi.org/10.48550/arXiv.2412.01506>

Xu, J. (2024). InstantMesh: Efficient 3D Mesh Generation from a Single Image with Sparse Views. <https://doi.org/10.48550/arXiv.2404.07191>